

Операционные системы

Продвинутое использование git

Крючков Дмитрий Алексеевич

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Получение навыков правильной работы с репозиториями git.

- 1 Выполнить работу для тестового репозитория.
- 2 Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

2. Процесс выполнения лабораторной работы

```
dakruchkov@dakruchkov:~$  
dakruchkov@dakruchkov:~$ pnpm setup  
Appended new lines to /home/dakruchkov/.bashrc  
  
Next configuration changes were made:  
export PNPM_HOME="/home/dakruchkov/.local/share/pnpm"  
case ":$PATH:" in  
  *:$PNPM_HOME:*) ;;  
  *) export PATH="$PNPM_HOME:$PATH" ;;  
esac  
  
To start using pnpm, run:  
source /home/dakruchkov/.bashrc  
dakruchkov@dakruchkov:~$ source ~/.bashrc  
dakruchkov@dakruchkov:~$
```

Рисунок 1: Node.js

```
dakruchkov@dakruchkov:~$ pnpm add -g commitizen

Update available! 10.27.0 → 11.25.0.
Changelog: https://pnpm.io/v/11.25.0
To update, run: pnpm self-update

WARN GET https://registry.npmjs.org/cli-cursor error (ECONNRESET). Will retry in 10 seconds. 2 retr
ies left.
WARN Request took 31614ms: https://registry.npmjs.org/cli-cursor
WARN GET https://registry.npmjs.org/wcwidth error (ECONNRESET). Will retry in 10 seconds. 2 retr
ies left.
WARN Request took 29389ms: https://registry.npmjs.org/wcwidth
Downloading @typescript/typescript-linux-x64@7.0.2: 9.67 MB/9.67 MB, done
WARN GET https://registry.npmjs.org/defaults/-/defaults-1.0.4.tgz error (ECONNRESET). Will retry in
10 seconds. 2 retries left.
WARN GET https://registry.npmjs.org/@typescript%2Ftypescript-linux-s390x error (ECONNRESET). Will r
etry in 10 seconds. 2 retries left.
WARN Request took 29121ms: https://registry.npmjs.org/@typescript%2Ftypescript-linux-s390x
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +148
+++++
Progress: resolved 167, reused 0, downloaded 148, added 148, done

/home/dakruchkov/.local/share/pnpm/global/5:
+ commitizen 4.3.2

Done in 1m 31.9s using pnpm v10.27.0
dakruchkov@dakruchkov:~$
```

Рисунок 2: установка commitizen


```

dakruchkov@dakruchkov:~$
dakruchkov@dakruchkov:~$ pnpm add -g standard-changelog
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +14
++++
Progress: resolved 181, reused 148, downloaded 14, added 14, done

/home/dakruchkov/.local/share/pnpm/global/5:
+ standard-changelog 8.1.0

Done in 1.8s using pnpm v10.27.0
dakruchkov@dakruchkov:~$
```

Рисунок 3: установка standard-changelog



Рисунок 4: package.json

```
da-kruhkov@da-kruhkov:~/git-extended$ git add .
da-kruhkov@da-kruhkov:~/git-extended$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:   Other changes that don't modify src or
test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
(4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 74c5f47] chore: test
1 file changed, 14 insertions(+)
create mode 100644 package.json
da-kruhkov@da-kruhkov:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 480 bytes | 480.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:da-kruhkov/git-extended.git
0ef79d1..74c5f47 master -> master
da-kruhkov@da-kruhkov:~/git-extended$
```

Рисунок 5: Отправка

```

dakruchkov@dakruchkov:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/dakruchkov/git-extended/.git/hooks]
dakruchkov@dakruchkov:~/git-extended$ git branch
* develop
  master
dakruchkov@dakruchkov:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/da-kruchkov/git-extended/pull/new/develop
remote:
To github.com:da-kruchkov/git-extended.git
 * [new branch]      develop -> develop
dakruchkov@dakruchkov:~/git-extended$
```

Рисунок 6: Инициализация и загрузка изменений

```
Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

dakruchkov@dakruchkov:~/git-extended$ standard-changelog --first-release
dakruchkov@dakruchkov:~/git-extended$ git add CHANGELOG.md
dakruchkov@dakruchkov:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 ff6fd2f] chore(site): add changelog
 1 file changed, 1 insertion(+)
 create mode 100644 CHANGELOG.md
dakruchkov@dakruchkov:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 CHANGELOG.md
Already on 'master'
Your branch is ahead of 'origin/master' by 2 commits.
(use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 CHANGELOG.md
Deleted branch release/1.0.0 (was ff6fd2f).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

dakruchkov@dakruchkov:~/git-extended$
```

Рисунок 7: Завершение релиза

Отправим данные на github

```
dakruchkov@dakruchkov:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 551 bytes | 551.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:da-kruchkov/git-extended.git
    74c5f47..f0ba387  develop -> develop
    74c5f47..dd09654  master -> master
dakruchkov@dakruchkov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 156 bytes | 156.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:da-kruchkov/git-extended.git
 * [new tag]          1.0.0 -> 1.0.0
dakruchkov@dakruchkov:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/da-kruchkov/git-extended/releases/tag/v1.0.0
dakruchkov@dakruchkov:~/git-extended$
```

Рисунок 8: Отправка

```
dakruchkov@dakruchkov:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

dakruchkov@dakruchkov:~/git-extended$ touch test2
dakruchkov@dakruchkov:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was f0ba387).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

dakruchkov@dakruchkov:~/git-extended$
```

Рисунок 9: Объединение веток

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

dakruchkov@dakruchkov:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 750 bytes | 750.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:da-kruchkov/git-extended.git
   f0ba387..4f619a4  develop -> develop
   dd09654..fd9ae60  master -> master
dakruchkov@dakruchkov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 157 bytes | 157.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:da-kruchkov/git-extended.git
 * [new tag]           1.2.3 -> 1.2.3
dakruchkov@dakruchkov:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/da-kruchkov/git-extended/releases/tag/v1.2.3
dakruchkov@dakruchkov:~/git-extended$
```

Рисунок 10: Завершение релиза


```

dakruchkov@dakruchkov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ gedit package.json
dakruchkov@dakruchkov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git add .
dakruchkov@dakruchkov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 1c0e1d1] chore: test
 1 file changed, 11 insertions(+), 11 deletions(-)
dakruchkov@dakruchkov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 476 bytes | 476.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:da-kruchkov/2026-1--study--os-intro.git
   a3c9cd8..1c0e1d1  master -> master
dakruchkov@dakruchkov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 11: package.json и коммит

```
Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

dakruchkov@dakruchkov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push -
-all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 690 bytes | 690.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To github.com:da-kruchkov/2026-1--study--os-intro.git
    1c0e1d1..8e7ffef  develop -> develop
    1c0e1d1..84cdf7  master -> master

dakruchkov@dakruchkov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push -
-tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 155 bytes | 155.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:da-kruchkov/2026-1--study--os-intro.git
 * [new tag]          1.0.0 -> 1.0.0
dakruchkov@dakruchkov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 12: Завершение релиза

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.